METHODS

Angane thanne, aadhyam #4 (vase volume) cheyyaam. Step-by-step ithaa:

Step 1: Real object edukkuka
Oru glass, flower vase allenkil kuzhal pole ulla pot edukkuka. Athinte **ullile (inner) radius** aanu venam, purathethalla. Ruler/scale vechu oro 1-2 cm height-ilum radius alakki ezhuthuka (cm-il). Ithu ninte real data aayi kaanikkaam.

Udaaharanam (ninte values vech maattanam):

Height (cm)	**0**	**2**	**4**	**6**	**8**	**10**
Radius (cm)	2.5	3.0	3.4	3.6	3.5	3.2

Step 2: GeoGebra-il points ittu curve fit cheyyuka
Input bar-il:

A=(0,2.5)

B=(2,3.0)

C=(4,3.4)

D=(6,3.6)

E=(8,3.5)

F=(10,3.2)

list={A,B,C,D,E,F}

f(x)=FitPoly(list,3)

H=10

Ippol vase-inte outline curve f(x) kittum. (Points nalla curve-il povunnillenkil degree 4 aakki nokkuka.)

Step 3: 3D-il solid undaakkuka
3D Graphics view open cheythu (View → 3D Graphics), height z-axis aakki:

vase=Surface(f(u) cos(v), f(u) sin(v), u, u, 0, H, v, 0, 2π)

Curve thirinju oru vase aayi maarunnathu kaanam.

Step 4: Volume calculate cheyyuka

V=π Integral(f(x)^2, 0, H)

Ithaanu disk method: $V = \pi \int [f(x)]^2 \, dx$.

Step 5: Disks kaanikkuka (judges-inu ishtappedum)
Oru slider n (1 to 50, increment 1) undaakki:

discs=Sequence(Cylinder((0,0,i H/n),(0,0,(i+1) H/n),f(i H/n)),i,0,n-1)

n koottumbol disks nirakkum vase-inte shape-ilekku varum, "integration = ethra cheriya disks koottunnathu" ennu visual aayi manassilaakkum. Disks-inte total volume-um text aayi kaanikkaam:

discVol=Sum(Sequence(π f(i H/n)^2 (H/n),i,0,n-1))

Step 6: Real check
Vase-il vellam nirachu measuring jug-il ozhichu ml-il alakkuka (1 ml = 1 cm³). Ennitt:

error=abs(V - realVolume)/realVolume*100

Text box-il "Calculated: __ cm³, Measured: __ cm³, Error: __%" ennu kaanikkuka.

Judges chodikkaan saadhyathayulla chodyangal

- *Disk method-il π r² enthina?* Oro cheriya slice-um oru circle aanu, athinte area π r², kanam dx koodumbol volume.
- *Calculated-um real-um thammil difference enthu kondu?* Glass-inte kanam, fitting error, alavile cheriya thettukal.
- *Ithu real life-il evide upayogikkunnu?* Bottle/tank design, pot manufacturing, engineering-il capacity kandupidikkaan.
- *Shell method use cheythaal?* Athum same answer tharum (extra marks-inu ready aakki vekkuka).

Tip: Real vase-um laptop-um pakshe kaanikkuka, GeoGebra file-il "before/after" pole real photo-yum insert cheyyaam (Image tool).

QUESTIONS FROM JUDGES

A. Project-inte basic chodyangal

1. **Ee project-il enthaanu cheythathu?**
   Oru real vase-inte inner radius pala height-ilum alannu, athil ninnu oru curve f(x) undaakki, athu axis-inu chuttum thirippichu 3D vase-inte volume integration vechu kandupidichu. Ennitt vellam ozhichu real volume alannu compare cheythu.
2. **Ee topic thiranjedukkaan karanam?**
   Integration textbook-il shape-inte area mathram aanu padikkunnathu. Athu real object-inte volume kandupidikkaan engane upayogikkaam ennu nokkaan aagrahichu.
3. **Ethu chapter-inte base?**
   Definite Integrals / Application of Integrals. Curve-inu keezhe ulla area kandupidikkunna athe aashayam 3D-ilekku extend cheythathaanu.

B. Maths-inte chodyangal

4. **V = π ∫ [f(x)]² dx ennu enthinaanu?**
   Vase-ine cheriya cheriya disks (cylinder pole) aayi murichaal, oro disk-inte radius f(x), kanam Δx aanu. Disk-inte volume = π r² × kanam = π [f(x)]² Δx. Ee disks ellaam kootti, Δx → 0 aakumbol sum integral aayi maarum.
5. **Disks-inte ennam (n) koodumbol enthu sambhavikkunnu?**
   Disk-inte sum (discVol) shariyaaya volume-ilekku aduthu varunnu. n = 1 aayappol 195.35 aayirunnu, n koodumbol 342.8-ilekku varunnu. Ithu tanne "definite integral oru sum-inte limit aanu" ennathinte visual proof.
6. **Ninte disk sum-um integral-um thammil difference undo?**
   Undu, chinna difference. Sum-il oro disk-inteyum kuzhal-inte thudakkathile radius mathrame

edukkunnullu, athu kondu n cheriyathaayaal thettu kooduthal. n valuthaakumbol athu kuranju integral value aayi thulyamaakum.

7. **f(x) engane kittiyathu? FitPoly enthaanu?**
   Alanna points (height, radius) ellaam kadannu pokunna, cubic polynomial (degree 3) aanu FitPoly kandupidikkunnathu. Least squares method aanu upayogikkunnathu, athayathu points-inodu ettavum aduthu nilkkunna curve.
8. **Enthinaanu degree 3 edutthathu? Degree 5 aayaal?**
   Vase-inte shape oru cheriya vakra maathram aayathu kondu cubic mathi. Degree koottiyaal curve points-ine kooduthal thottu pokum, pakshe angane pokumbol points-inte idayil curve kurachu aadi (oscillate) povaan saadhyatha und, athu volume-ine thettikkum.
9. **Verum oru curve thirippichaal ee shape kittumo? Vase-inu circular symmetry venamo?**
   Venam. Ee method-inte assumption vase axis-inu chuttum ellaa dikkilum ore radius aanu ennathaanu. Oval allenkil chathuram vase-ine ithu vachu alakkaan pattilla.

C. Real experiment-inte chodyangal

10. **Calculated-um measured-um thammil difference enthu kondu?**

1. Glass-inte kanam, athayathu inner-um outer-um radius thammilulla vyathyaasam.
2. Ruler kondulla alavil ulla cheriya thettu.
3. 6 points mathram edutthathu (kooduthal points edutthaal nannaakum).
4. FitPoly curve-inte cheriya fitting error.
5. Vase-inte thaazhe/mukalile vakra bhaagam perfect circle allathathu.

11. **Enthinaanu inner radius alannathu?**
    Vellam nirakkunnathu ullilaanu, athu kondu inner volume aanu venam. Purathethu alannaal glass-inte kanam koodi ulppedum, volume kooduthal varum.
12. **1 ml = 1 cm^3 ennu engane ariyaam?**
    1 litre = 1000 cm^3 aanu, athu kondu 1 ml = 1 cm^3. Athaanu cm-il alannathu.
13. **Error % eth vare accept cheyyaam?**
    Real experiment-il 5-10% vare sadharanam aanu. Athilum kooduthal aanenkil measurements onnukoodi check cheyyanam. Ninte error ethra vannu ennu clear aayi parayuka.

D. Extra / bonus chodyangal

14. **Ithe shell method vechu cheythaal?**
    Shell method-il vase-ine kanam kuranja cylinder-pole ulla shells aayi murikkum, $V = 2\pi \int (\text{radius})(\text{height})\, d(\text{radius})$. Randu method-um ore answer tharum, pakshe ivide disk method aanu ealuppam, karanam radius height-inte function aayi ippol kittiyittundu.
15. **Ithu real life-il evide upayogikkunnu?**
    Bottles, tanks, pots, vases-inte capacity design cheyyaan, water tank-il ethra litre undu ennu alakkaan, engineering-il rotating parts (shaft, turbine) design cheyyaan.
16. **Ithu improve cheyyaan enthu cheyyaam?**
    Kooduthal points edukkuka (ee 6 points-ilum kooduthal), 3D scanner allenkil photo-yil ninnu digitize cheyyuka, vevveru vases-il try cheythu compare cheyyuka.

17. GeoGebra illathe ithu cheyyaan pattumo?
Pattum, kaiyil polynomial fit cheythu integral solve cheyyaam. Pakshe GeoGebra fitting, integration, 3D visualization ellaam vegam cheyyunnu, athu kondu upayogichu.

METHODS

Upayogicha saamagrikal (Materials)

- Oru real vase / glass (vrithaakruthiyil ullathu)
- Scale / vernier caliper
- Measuring jug, vellam
- Laptop, GeoGebra

Cheytha reethi (Procedure)

1. **Alavedukkal:** Vase-inte inner radius adiyil ninnu oro 2 cm height-ilum alannu, pattika aakki (height, radius).
2. **Points idal:** Aa alavukal GeoGebra-yil points aayi (A, B, C...) koduthu.
3. **Curve fit cheyyal:** FitPoly(list, 3) upayogichu ellaa points-inum ettavum aduthulla cubic function f(x) kandupidichu.
4. **3D vase undaakkal:** Aa curve-ine z-axis-inu chuttum 0 muthal 2π vare thirichu Surface command kondu 3D roopam undaakki.
5. **Volume kanakkaakkal:** $V = \pi \int_0^H [f(x)]^2 \, dx$ formula upayogichu volume kandupidichu.
6. **Disk sum kaanikkal:** Slider n upayogichu vase-ine n disks aayi murichu, avayude volumes-inte sum (discVol) kanakkaakki. n koodumbol ithu integral value-ilekku aduthu varunnathu nireekshichu.
7. **Real check:** Vase-il vellam nirachu measuring jug-ilekku ozhichu ml-il alannu (1 ml = 1 cm^3).
8. **Compare cheyyal:** error % = |V − measured volume| / measured volume × 100 kanakkaakki, difference-inte kaaranangal vishakalanam cheythu.

Upayogicha maths reethikal (Mathematical Methods)

Reethi	Enthinu
Least squares curve fitting (FitPoly)	Alavukalil ninnu function kandupidikkaan
Solid of revolution, disk method	Vase-inte volume kanakkaakkaan
Definite integral	$V = \pi \int [f(x)]^2 \, dx$ kanakkaakkaan
Riemann sum (disk sum)	Integral oru sum-inte limit aanennu kaanikkaan
Percentage error	Calculated-um real-um compare cheyyaan

Presentation-inu oru vari
"Alakkuka → points → curve → 3D vase → integral → disk sum → vellam kondu check → error." Ee kramathil flowchart aayi chart paper-il varachaal judges-inu pettannu manassilaakum.